

Name: _____ Date: _____

Period: _____

Parametric Rates of Change Practice

1. A particle moves with $h(t) = (t^2 - 4t, 2t - t^2 + 1)$ for $0 \leq t \leq 4$.

(a) Complete the table.

t	$x(t) = t^2 - 4t$	$y(t) = 2t - t^2 + 1$	(x, y)
0	_____	_____	_____
1	_____	_____	_____
2	_____	_____	_____
3	_____	_____	_____
4	_____	_____	_____

(b) At $t = 0.5$, is the particle moving right or left? Up or down? Show your reasoning by comparing table values.

(c) At $t = 3$, is the particle moving right or left? Up or down?

Problems 2–3: Average Rates of Change and Slope

2. Use $h(t) = (t^2 - 4t, 2t - t^2 + 1)$ from Problem 1.

(a) Compute the slope of the curve between $t = 0$ and $t = 2$. Show the computation of Δx and Δy separately, then form the ratio.

(b) Compute the slope of the curve between $t = 1$ and $t = 4$.

3. A different particle moves with $p(t) = (3t - 6, t^2 - 4t + 3)$ for $0 \leq t \leq 5$.

(a) Compute the slope of the curve between $t = 1$ and $t = 3$.

(b) At what value of t in $[0, 5]$ is $x(t) = 0$? State the particle's position at that moment.

Problem 4: Two Parametrizations of the Same Line

4. Consider the straight-line path traced by $f(t) = (2t, 3t - 1)$ for $0 \leq t \leq 3$.

(a) State the start point and end point of f .

(b) Write a second parametric function $g(t)$ that traces the *same* straight-line path from the end point back to the start point as t increases from 0 to 3. (Hint: reverse the direction.)

(c) Compute the slope of the curve for f between $t = 0$ and $t = 3$. Then compute the slope for g between $t = 0$ and $t = 3$ (which are the reversed endpoints). Are the slopes equal? Explain.

Problem 5: AP-Style Multiple Choice

5. A particle moves in the plane with parametric function $f(t) = (t^2 - 2t, t^2 + 2t)$ for $0 \leq t \leq 4$. Which of the following correctly describes the particle's motion at $t = 1$?

- (A) Moving right and up
- (B) Moving left and up
- (C) Moving left and down
- (D) Moving right and down

Show your work:
